

Anatomy & Kinesiology Wrist & Hand

Bones and Joints

- **Distal radius and ulna:**
 - *Ulnar tilt* 25° — reason why ulnar deviation > radial deviation.
 - *Palmar tilt* 10° — reason why wrist flexion > wrist extension.
- **Carpal bones:**
 - **Proximal row:** Scaphoid, Lunate, Triquetrum, Pisiform
 - **Distal row:** Trapezium, Trapezoid, Capitate, Hamate
 - **Other terms:**
 - Scaphoid = navicular
 - Trapezium = greater multangular
 - Trapezoid = lesser multangular
 - Triquetrum = triangular
 - Capitate = Os magnum
 - **Lister's tubercle:** dorsal tubercle separating tendons of EPL and ECRB.

Wrist Ligaments

- **Extrinsic ligaments:**
 - Dorsal radiocarpal
 - Radial collateral
 - Palmar radiocarpal
 - Radioscapholunate
 - Radiolunate
 - Radioscaphocapitate
 - **Triangular Fibrocartilage Complex (TFCC):**
 - Primary stabilizer of distal RU joint
 - Reinforces ulnar side of wrist
 - Helps transfer compressive forces from hand to forearm
 - RU joint capsular ligament
 - Palmar ulnocarpal (ulnotriquetral, ulnolunate)
 - Ulnar collateral ligament
 - Meniscus homologue
- **Intrinsic ligaments:**
 - Short (distal row): dorsal, palmar, interosseous
 - Intermediate:
 - Lunotriquetral

- Scapholunate
- Scaphotrapezi al
- Long:
 - Palmar intercarpal (inverted V):
 - Lateral leg (capitate to scaphoid)
 - Medial leg (capitate to triquetrum)
 - Dorsal intercarpal (trapezium, scaphoid, lunate, triquetrum)

Wrist Kinematics

- RC motion: greater in flexion (50°), MC motion (35° flexion)
- MC motion: greater in extension (50°), RC (35° extension)
- **Radial deviation:**
 - Roll: lateral
 - Glide: medial
 - Ligaments taut: ulnar collateral, palmar radiocarpal (RC), palmar intercarpal medial leg (MC)
- **Ulnar deviation:**
 - Roll: medial
 - Glide: lateral
 - Ligaments taut: radial collateral, palmar ulnocarpal (RC), palmar intercarpal lateral leg (MC)

Metacarpals (5)

- 2nd MCP abduction: roll + glide lateral
- 3rd MCP joint = axis
- 4th/5th MCP abduction: roll + glide medial
- **Deep transverse ligament:** prevents splay deformity

THUMB CARPOMETACARPAL (CMC) JOINT

The ligaments of the thumb CMC joint are located on all sides, contributing to both stability and mobility. Their naming can be confusing due to different classification systems.

The anterior oblique ligament (beak ligament) connects the palmar tubercle of the first metacarpal base to the trapezium. It is the primary stabilizing ligament of the joint and becomes taut during abduction, extension, and opposition. Some authors describe it as having superficial and deep components.

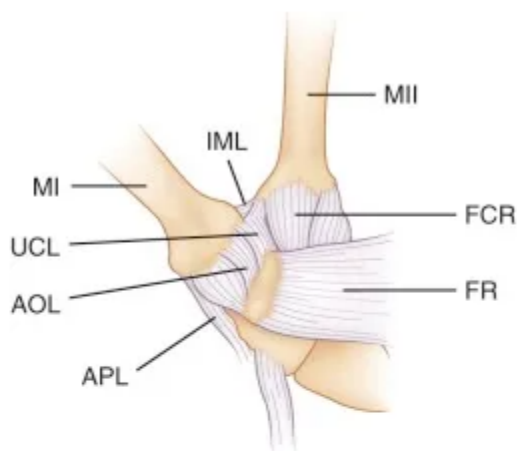
The ulnar collateral ligament is found on the volar and medial side, extending from the transverse carpal ligament to the palmar-medial base of the first metacarpal. It provides medial stability.

The posterior oblique ligament lies dorsally and connects the trapezium to the medial base of the first metacarpal.

The dorsoradial ligament extends from the dorsolateral trapezium to the dorsal base of the first metacarpal and is an important dorsal stabilizer.

The intermetacarpal ligament connects the bases of the first and second metacarpals and adds additional support.

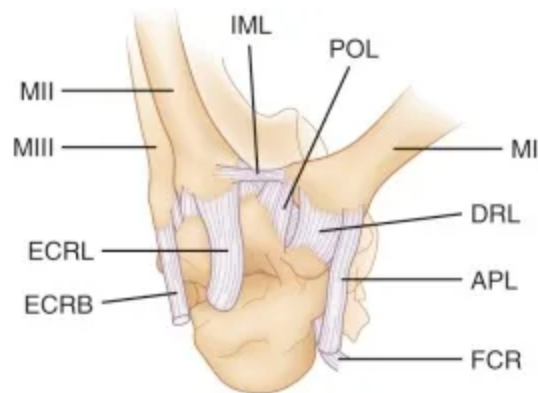
The joint capsule is complete but relatively loose, allowing the thumb its essential axial rotation for opposition.



Key	
AOL,	Anterior oblique ligament
UCL,	Ulnar collateral ligament
IML,	First intermetacarpal ligament
APL,	Abductor pollicis longus tendon
FCR,	Flexor carpi radialis
FR,	Flexor retinaculum
MI,	First metacarpal
MII,	Second metacarpal

Figure 1-6

Palmar view of the ligaments of the carpometacarpal joint of the left thumb.



Key	
IML,	First intermetacarpal ligament
POL,	Posterior oblique ligament
DRL,	Dorsoradial ligament
APL,	Abductor pollicis longus tendon
ECRL,	Extensor carpi radialis longus tendon
ECRB,	Extensor carpi radialis brevis tendon
MIII,	Third metacarpal

Figure 1-7

Dorsal view of the ligaments of the carpometacarpal joint of the left thumb.

METACARPOPHALANGEAL (MCP) JOINTS OF DIGITS 2–5

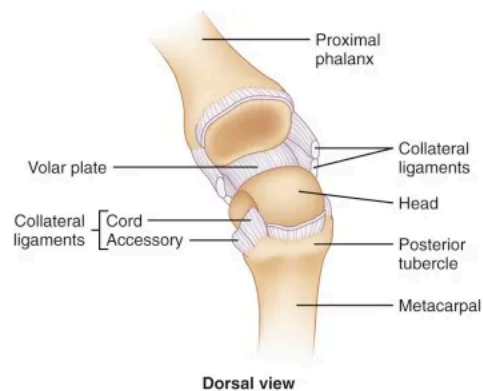


Figure 1-8

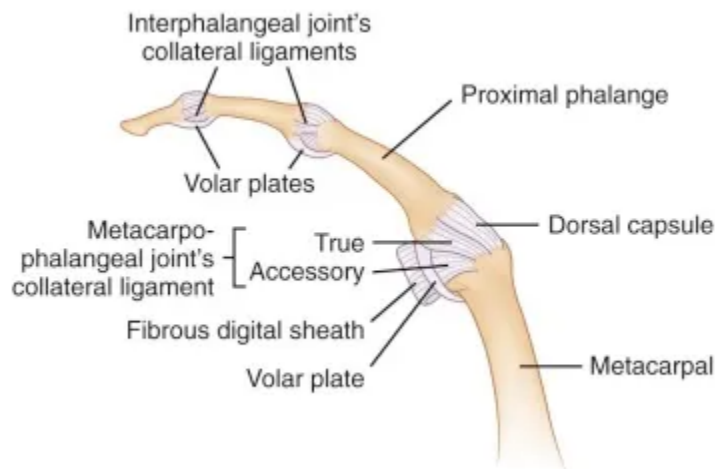
Dorsal view of the metacarpophalangeal joint that is opened dorsally to show the articular surfaces. Note the biconvex metacarpal head and the biconcave proximal phalangeal base.

The MCP joints are formed by the convex heads of the metacarpals and the concave bases of the proximal phalanges.

The metacarpal head is biconvex and cam-shaped, extending further volarly and being wider on the volar side. The base of the proximal phalanx is biconcave and smaller. Although this shape allows potential movement in multiple planes, actual motion is limited by soft tissues.

The MCP joints allow flexion, extension, abduction, and adduction. Rotation is only possible passively. Abduction and adduction occur relative to the middle finger.

JOINT CAPSULE AND VOLAR PLATE



The MCP joint capsule encloses the joint and is specialized on the volar side by the volar plate, a fibrocartilaginous structure.

The volar plate prevents hyperextension and provides attachment for the A1 pulley of the flexor tendon sheath, helping maintain proper positioning of the flexor tendons.

Distally, the volar plate is thick and firmly attached to the proximal phalanx. Proximally, it is thinner and more flexible, attaching to the metacarpal head. During flexion, it slides proximally and folds, allowing movement.

COLLATERAL LIGAMENTS OF MCP

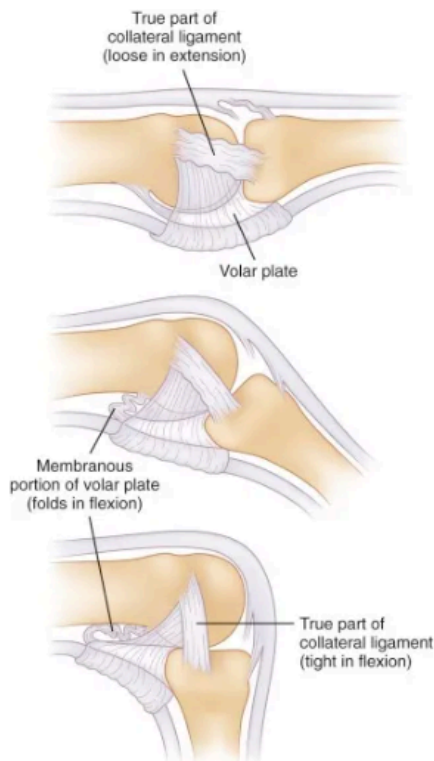


Figure 1-10

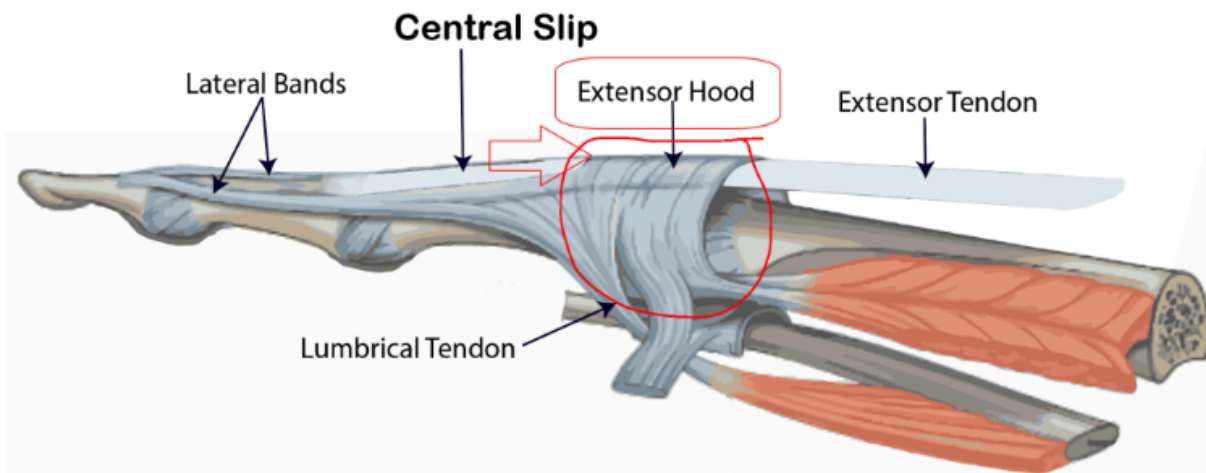
Lateral view of the metacarpophalangeal joint of a finger. The band part of the collateral ligament and the volar plate are depicted in full extension, partial flexion, and flexion. Note how the tension of the band part of the ligament changes as the proximal phalanx is flexed. Note also how the proximal part of the volar plate folds as flexion occurs.

The collateral ligament has two parts.

The true (proper) collateral ligament runs obliquely from the dorsal metacarpal tubercle to the volar aspect of the proximal phalanx. It is loose in extension, allowing abduction and adduction, and becomes tight in flexion due to the cam shape of the metacarpal head, limiting side-to-side motion.

The accessory collateral ligament attaches to the volar plate and helps stabilize both the joint and the flexor tendons. It loosens slightly during flexion.

EXTENSOR HOOD



The MCP joints are reinforced by the extensor hood, a fibrous structure that surrounds the joint.

It includes sagittal bands that keep the extensor tendon centered. The hood transmits force from the extensor tendons to extend the proximal phalanx and moves proximally and distally during motion.

THUMB MCP JOINT

The thumb MCP joint is similar to the other MCP joints but has unique features.

The volar plate contains two sesamoid bones that create a groove for the flexor pollicis longus tendon.

The ulnar side of the extensor hood is stronger and formed by the adductor pollicis, while the radial side is formed by the abductor pollicis brevis and flexor pollicis brevis.

This joint is more stable than the MCP joints of the fingers, so its motion is more limited. However, it is more vulnerable to abduction and adduction forces, making it prone to injury such as ulnar collateral ligament injury.

- **Arthrokinematics:**

- Abduction: roll anterior, glide posterior
- Flexion: roll + glide laterally
- Opposition:
 - Phase 1: roll anterior, glide posterior

- Phase 2: roll + glide medially

PROXIMAL INTERPHALANGEAL (PIP) JOINT

The PIP joint is formed by the head of the proximal phalanx and the base of the middle phalanx.

The head is cylindrical, and the base is concave, with a sagittal groove and ridge that enhance stability. This design restricts movement to a single plane.

The PIP joint allows only flexion and extension.

PIP JOINT SUPPORTING STRUCTURES

The volar plate is similar to that of the MCP joint and prevents hyperextension. It includes check-rein ligaments, which tighten during extension to limit excessive motion. It also serves as an attachment for the A3 pulley.

The collateral ligaments have true and accessory components. The true collateral ligament remains tight throughout the range of motion, providing constant stability, while the accessory ligament supports the volar plate.

Dorsally, the joint is reinforced by the extensor mechanism, including the central band and triangular ligament. Laterally, the lateral bands and retinacular ligaments provide support.

Both the flexor digitorum superficialis and profundus tendons pass volarly across the joint.

DISTAL INTERPHALANGEAL (DIP) JOINT

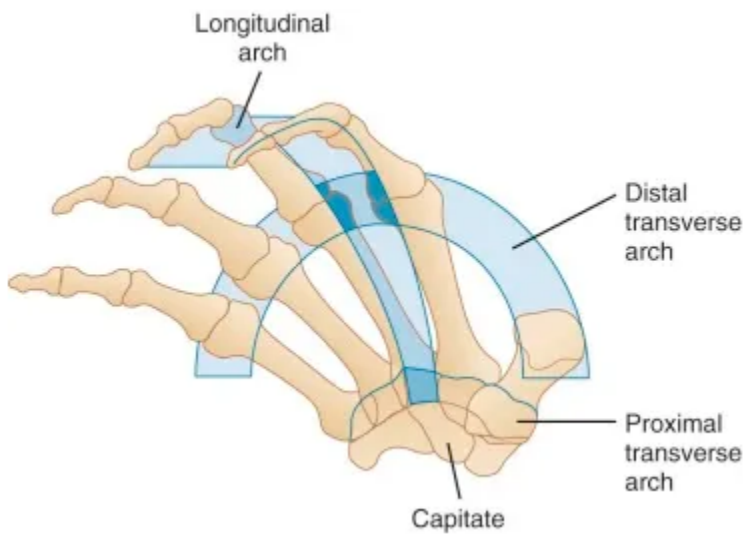
The DIP joint is structurally similar to the PIP joint but simpler.

It allows only flexion and extension due to its hinge-like structure.

The volar plate prevents hyperextension and provides attachment for the A5 pulley.

Only the flexor digitorum profundus tendon crosses the volar side, and dorsally the extensor mechanism is less complex, primarily involving the terminal tendon.

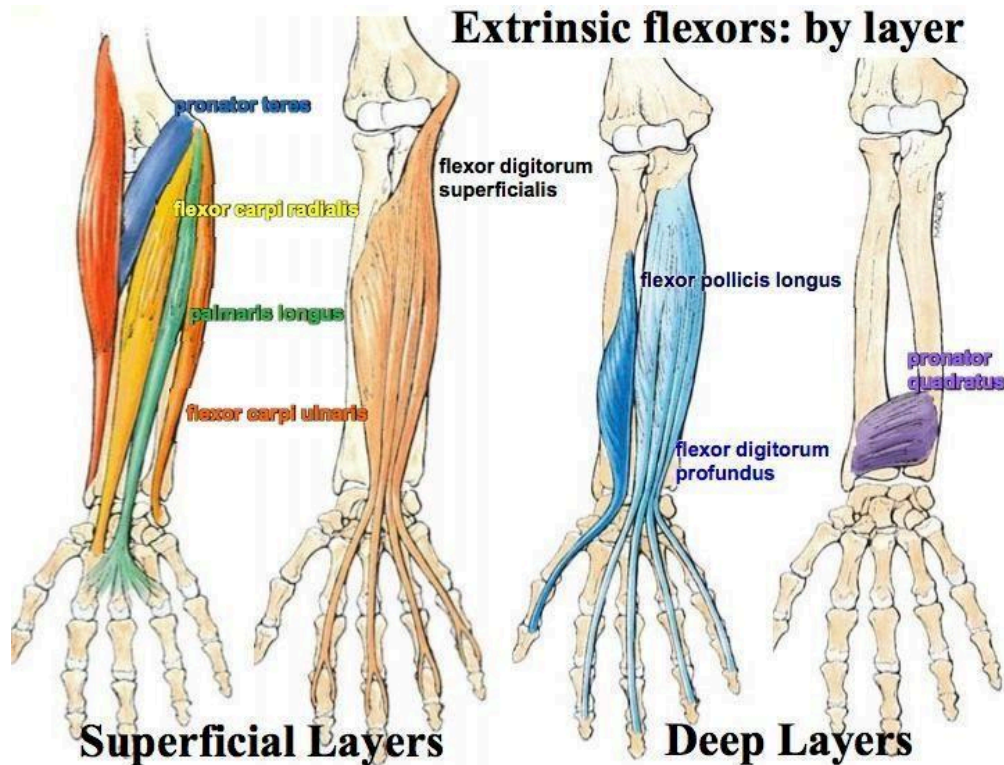
Arches of the Hand



- Proximal transverse arch: keystone = capitate
- Distal transverse arch: keystone = middle finger
- Longitudinal arch: keystone between 2nd/3rd finger

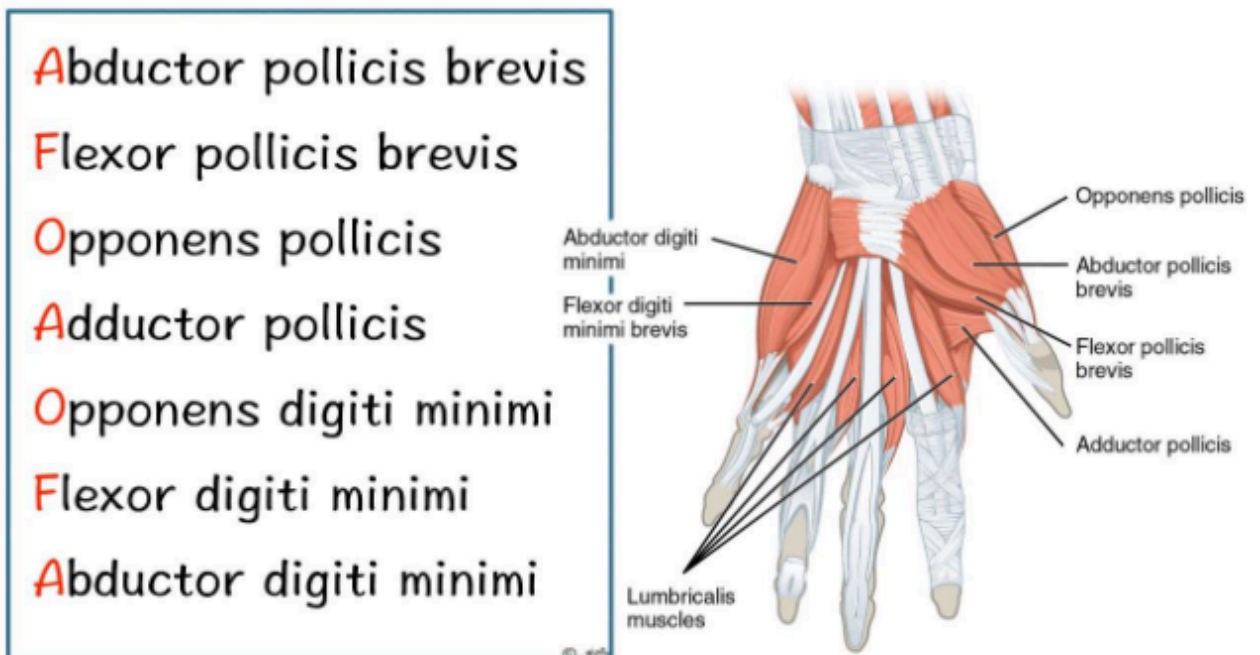
Muscles of Wrist & Hand

- **Extrinsics:**
 - *Anterior compartment (superficial):* FCR, PL, FCU (ulnar nerve)
 - *Intermediate:* FDS
 - *Deep:* FPL, FDP (flex DIP, dually innervated; only muscle active in unresisted closure)
 - *Lateral compartment (radial nerve):* Brachioradialis, ECRL
 - *Posterior superficial:* ECRB, ED, EDM, ECU (wrist extensors increase contraction with stronger grip)
 - *Posterior deep:* APL, EPB, EPL, EIP



- Intrinsic:

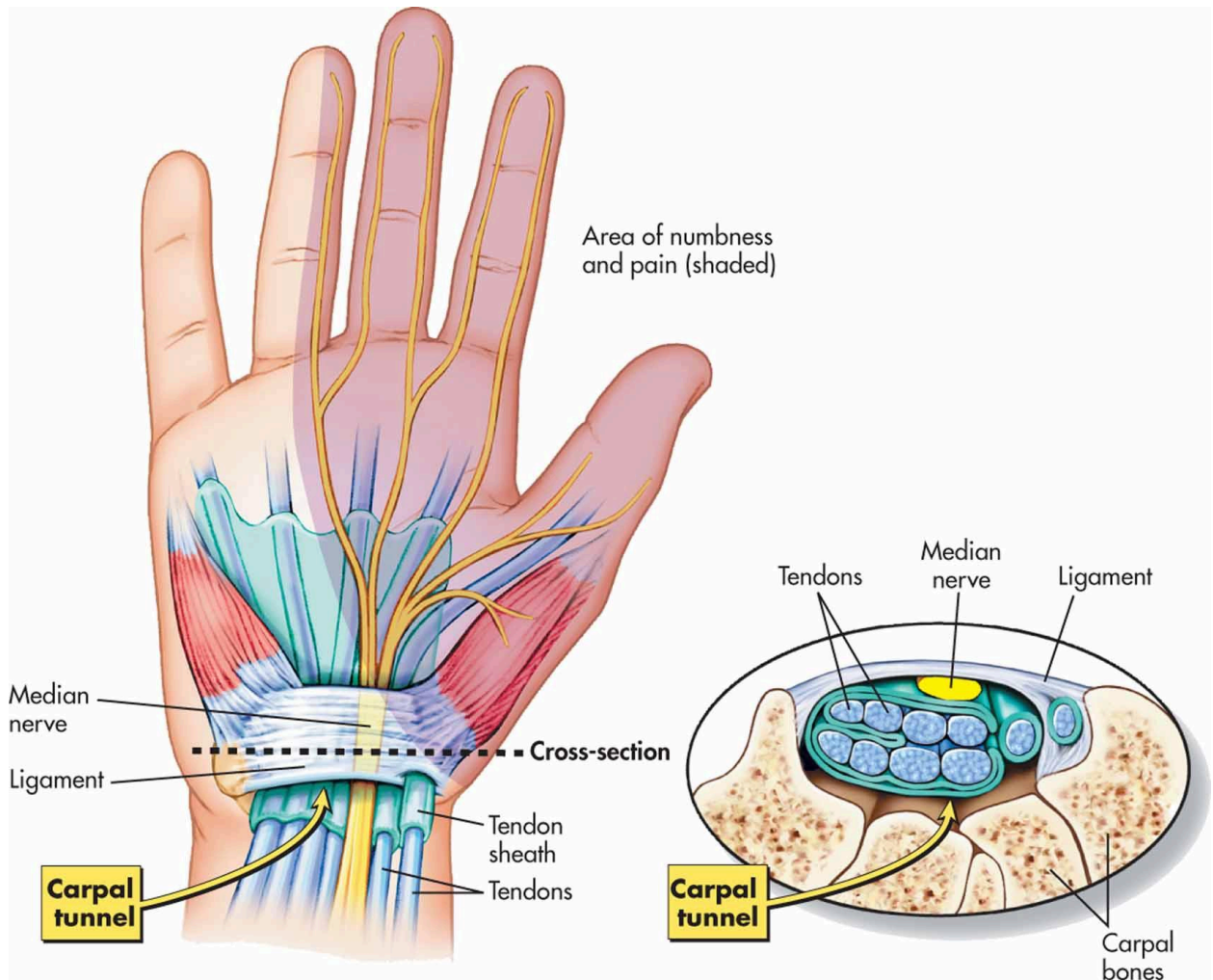
“All For One And One For All”



- *Small muscles*: Palmaris brevis (improves grip); lumbricals (silent in MCP flexion if IP joints flexed)
- *Interossei*: stabilize extensor hood, pinch, grasp
- *Adductor pollicis*
- *Thenar*: APB, FPB, Opponens pollicis
- *Hypothenar*: Opponens digiti minimi, FDM, Abductor digiti minimi

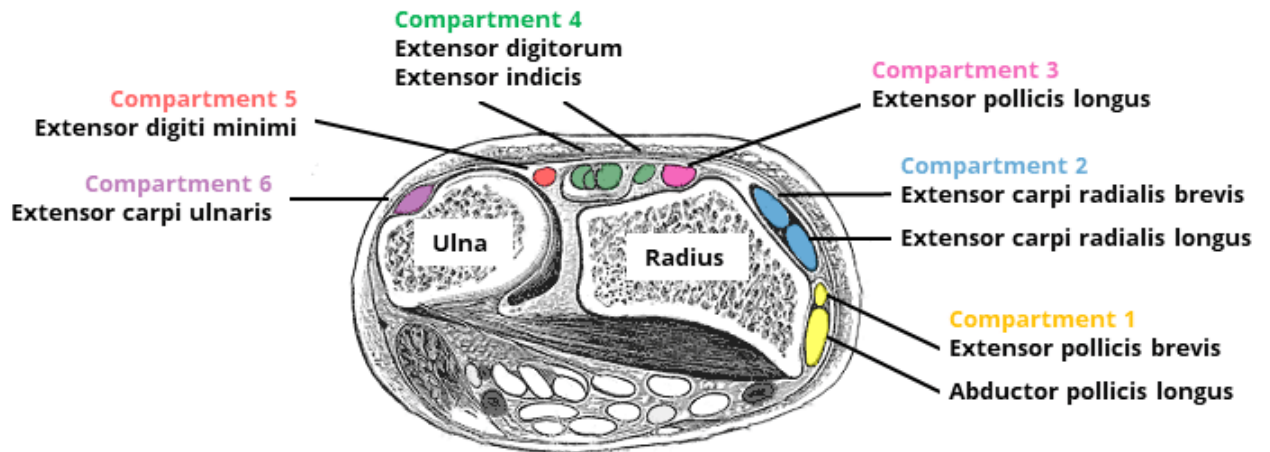
Special Topics

- **Carpal Tunnel:**



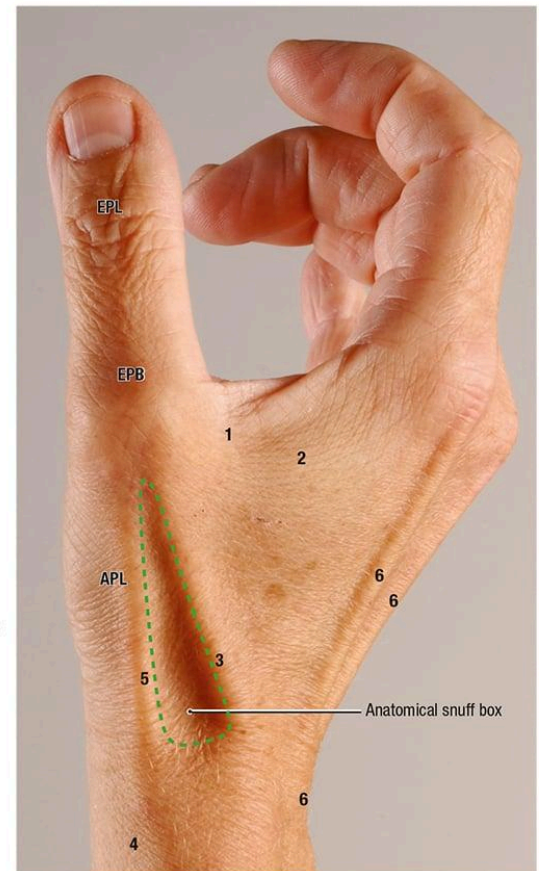
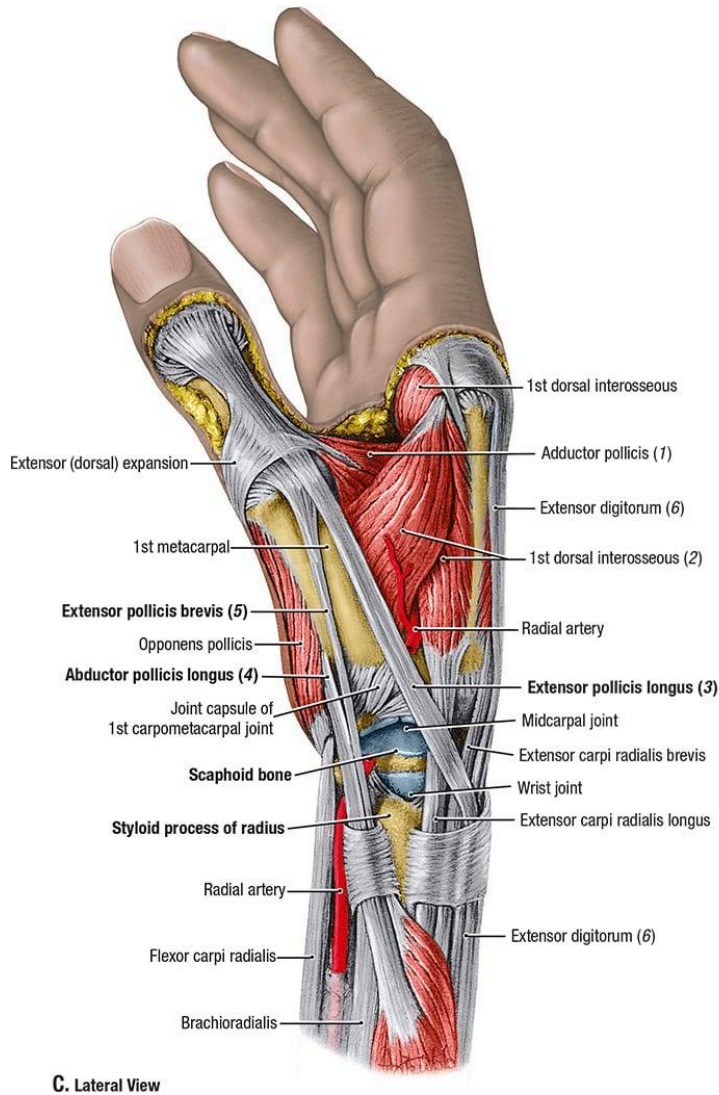
- Roof: transverse carpal ligament (pisiform & hook of hamate to scaphoid & trapezium)
- Contents: 4 FDP, 4 FDS, median nerve, FPL

- **Extensor Tunnel:**



- Attachments: pisiform & hook of hamate to distal radius
- Contents:
 - 1st: APL, EPB
 - 2nd: ECRL, ECRB
 - 3rd: EPL
 - 4th: EDC, EIP
 - 5th: EDM
 - 6th: ECU

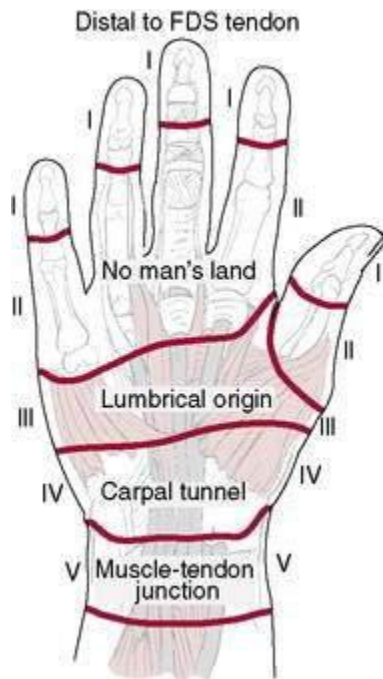
- **Anatomic snuffbox:**



Distal Extent of	
APL	Abductor pollicis longus
EPB	Extensor pollicis brevis
EPL	Extensor pollicis longus

- Borders: APL/EPB (radial), EPL (ulnar), floor = scaphoid/trapezium, content = radial artery

- **No man's land:**

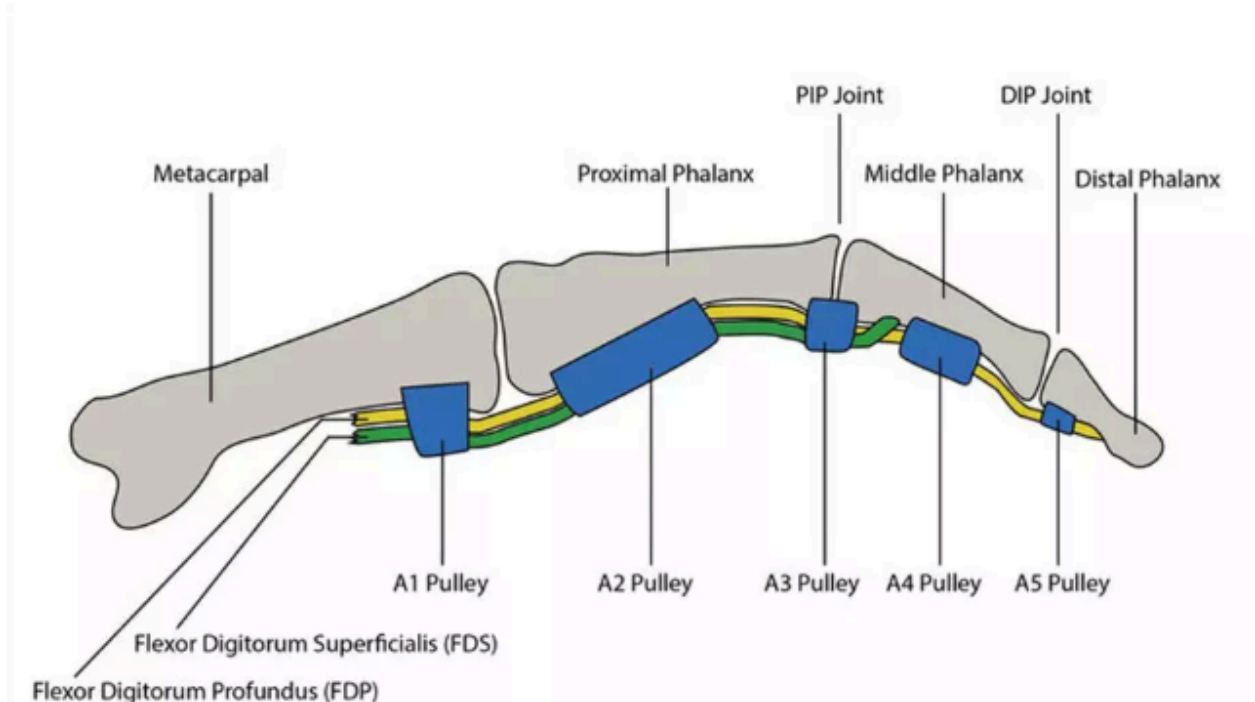


- Between distal palmar crease and PIP crease; tendon repair is contraindicated

- **Extensor assembly:**

- Tendinous: extrinsic extensors + intrinsic muscles
- Retinacular: fascia + ligaments
- Components:
 - Central band — transmits ED force across IP
 - Lateral bands — transmits force across PIP/DIP
 - Dorsal hood (transverse & oblique fibers)
 - Oblique retinacular ligament — coordinates PIP & DIP motion

- **Flexor Pulleys:**



The flexor pulley system is a series of fibrous bands that hold the flexor tendons (flexor digitorum superficialis and flexor digitorum profundus) close to the bones of the fingers.

Its main function is to prevent bowstringing during finger flexion and to ensure efficient force transmission from muscle to bone.

GENERAL FUNCTION

When the flexor tendons contract, they tend to pull away from the bone. The pulleys act like restraints that keep the tendons close to the phalanges, allowing smooth and efficient flexion.

Without pulleys, finger flexion would be weak and mechanically inefficient.

CLASSIFICATION OF PULLEYS

There are two types of pulleys in each finger:

1. Annular pulleys (A pulleys)
2. Cruciform pulleys (C pulleys)

ANNULAR PULLEYS (A1–A5)

These are strong, thick bands and are the most important for preventing bowstringing.

A1 pulley

Located at the level of the metacarpophalangeal (MCP) joint

Attached to the volar plate of the MCP joint

Common site of trigger finger

A2 pulley

Located along the proximal phalanx

Considered the most important pulley for preventing bowstringing

Maintains optimal tendon biomechanics

A3 pulley

Located at the proximal interphalangeal (PIP) joint

Attached to the volar plate of the PIP joint

A4 pulley

Located along the middle phalanx

Works with A2 to maintain tendon efficiency

A5 pulley

Located at the distal interphalangeal (DIP) joint

Attached to the volar plate of the DIP joint

CRUCIFORM PULLEYS (C1–C3)

These are thinner and more flexible.

They are located between the annular pulleys and allow the tendon sheath to collapse and expand during finger movement.

Their function is to provide flexibility and prevent stiffness or constriction of the tendon sheath during flexion.

BIOMECHANICAL IMPORTANCE

The A2 and A4 pulleys are the most critical for normal finger function.

If these pulleys are damaged, bowstringing occurs, where the tendon pulls away from the bone, resulting in reduced grip strength and inefficient motion.

CLINICAL CORRELATIONS

Trigger finger

Caused by thickening or inflammation of the A1 pulley

Leads to catching or locking of the finger during movement

Pulley rupture

Common in rock climbers due to excessive load

Most commonly affects A2 pulley

Leads to visible bowstringing and weakness

Flexor tendon injury

Pulley integrity must be preserved during repair to maintain function

Power & Precision Grips

Grip	Description	Finger Function	Thumb Function	Example
Lateral pinch (key)	Object between index finger & thumb	Index abducted, partially flexed MCP/IP; FDP, FDS, 1st dorsal interossei active	Thumb abducted, IP flexed; FPL, FPB, adductor pollicis	Holding keys, grasping paper
Three-prong chuck	Pads of 2nd, 3rd digits & thumb	MCP/PIP flexion; FDS (if DIP extended), FDP (if flexed)	Thumb in opposition, flexed MCP/IP, adduction at CMC; adductor pollicis stabilizes	Holding coins, writing
Tip-to-tip grip	Tip prehension between thumb & finger	Full flexion of fingers; FDP primary mover	Thumb in opposition, flexed MCP/IP; FPL important	Picking up small items
Hook grip	Used for carrying objects	FDS & FDP active, IP flexion	Thumb not involved	Carrying bags, weights
Cylinder grip	Holding cylindrical objects	Flexion of MCP/IP; interossei counterbalance radial ligaments	Thumb flexed/adducted, opposing fingers	Holding bottles, phone
Spherical grip	Holding round objects	MCP abduction, partial flexion; extrinsic flexors, ED co-contract	Thumb opposition, MCP/IP flexion, thenar muscles active	Holding a ball or apple
Fist grip	Thumb wraps around object	Flexion of all finger joints	Thumb adducted or opposed to fingers	Gripping small tools, bat

Power Grip



Figure 7-39 Types of power grips.

Precision Grip

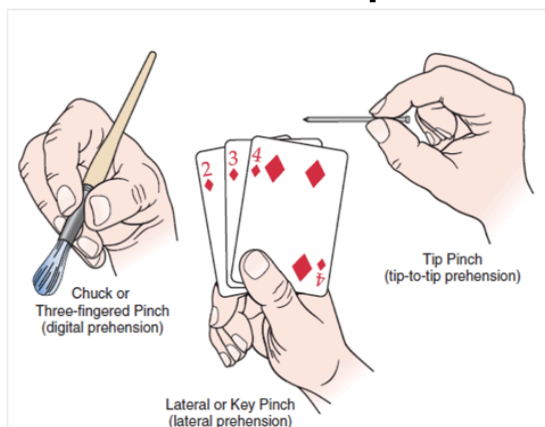


Figure 7-40 Types of precision grips or pinches.

